

- ① The Company wants to automate its payroll process. Write a C program to calculate the total Salary by adding the basic Salary and bonus amount of an employee.

PSEUDO CODE:

START

READ basic-Salary

READ bonus

total-Salary = basic-Salary + bonus

PRINT total-Salary

STOP

INPUT:

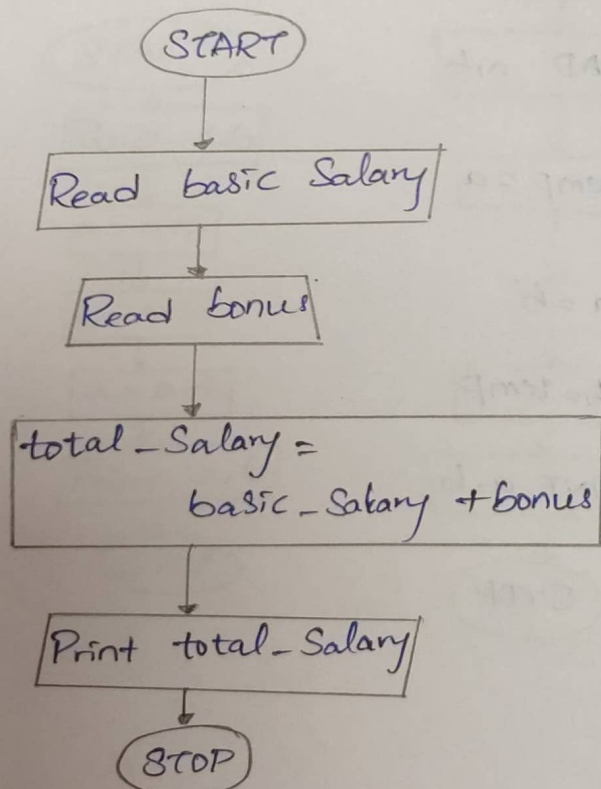
basic-Salary = 30000

bonus = 5000

OUTPUT:

total Salary = 35000

FLOW CHART



② During a database update, two employee IDs used needs to be exchanged. Write a C program to Swap two numbers using a temporary Variable.

PSEUDO CODE :

```
START
  READ a, b
  temp = a
  a = b
  b = temp
  PRINT a, b
STOP
```

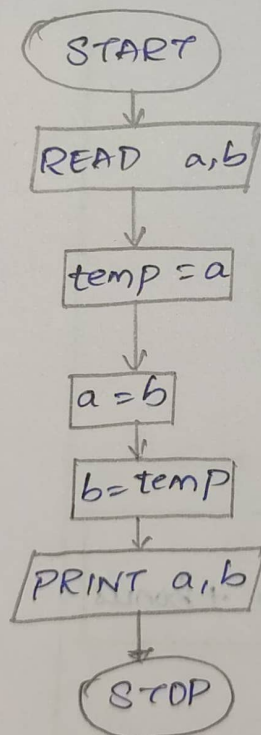
INPUT:

a=5, b=10

OUTPUT:

a=10, b=5

FLOW CHART:



③

In an Embedded System with limited memory, two values must be exchanged without using any extra storage. Write a C program to swap two numbers without using a temporary variable.

PSEUDO CODE:

START

READ a,b

$a = a + b$

$b = a - b$

$a = a - b$

PRINT a,b

STOP

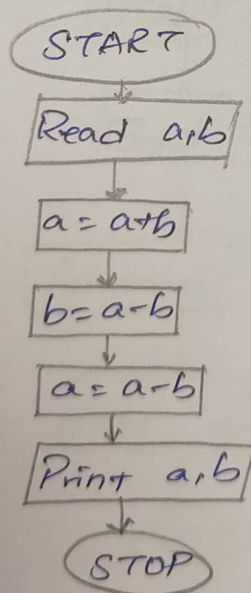
INPUT:

$a = 5, b = 10$

OUTPUT:

$a = 10, b = 5$

FLOW CHART:



- ④ A Bank. wants to provide customers with an estimate of the interest earned on their deposits. Write a C program to calculate the Simple Interest using formula $SI = (P \times R \times T) / 100$.

PSEUDO CODE:

START

READ P, R, T

$SI = (P \times R \times T) / 100$

PRINT SI

STOP

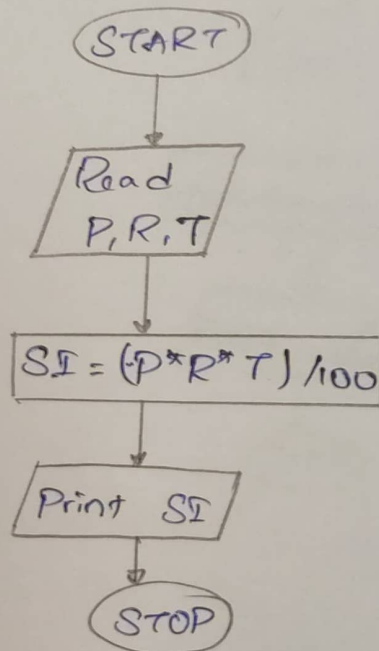
INPUT:

$P = 10000, R = 5, T = 2$

OUTPUT:

Simple Interest = 1000

FLOW CHART:



⑤

A weather station records temperature in Celsius, but reports need to be generated in Fahrenheit. Write a C program to convert temp from Celsius to Fahrenheit.

PSEUDO CODE:

START

READ Celsius

$fahrenheit = (Celsius * 9/5) + 32$

PRINT fahrenheit

STOP

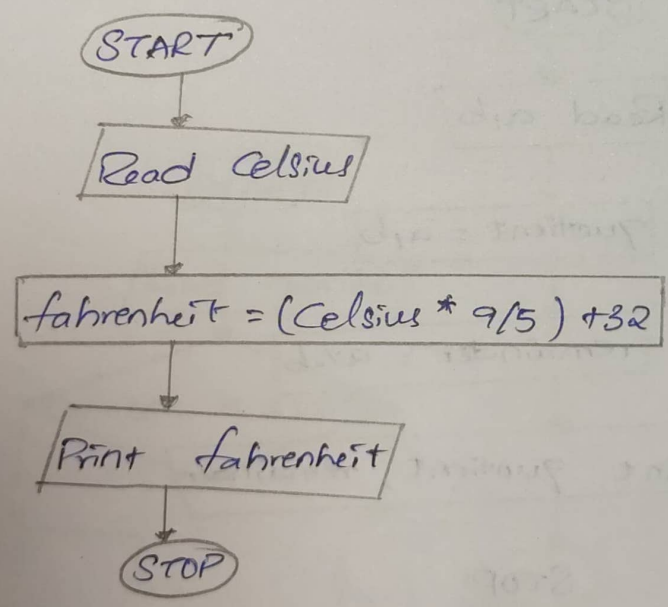
INPUT:

Celsius = 37

OUTPUT:

Fahrenheit = 98.6

FLOW CHART:



- ⑥ A Mathematics learning application needs to be display both the quotient and remainder when two numbers are divided write a C program to find the quotient & remainder of two integers.

PSEUDO CODE:

START

READ a, b

quotient = a/b

remainder = $a \% b$

PRINT quotient, remainder

STOP

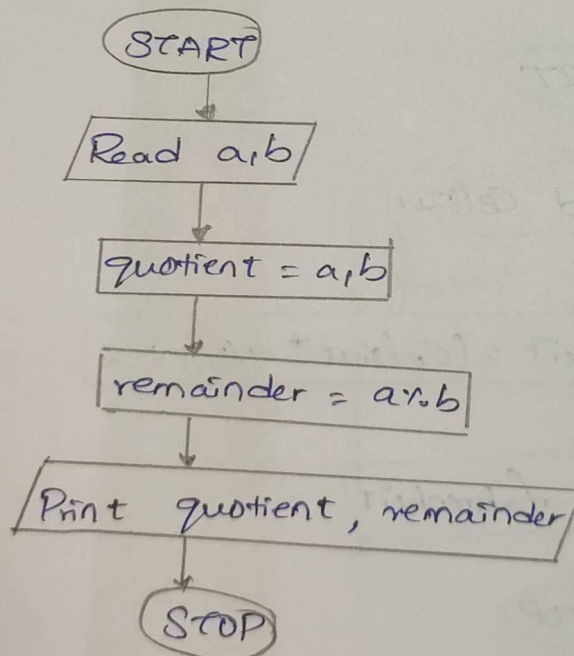
INPUT:

$a=17, b=5$

OUTPUT:

quotient = 3, Remainder = 2.

FLOW CHART:



⑦ An Event Management System allows entry based on even numbered ticket IDs. Write a C program to determine whether a given ticket number is even or odd.

PSEUDO CODE

START

READ number

IF number % 2 == 0 THEN

PRINT "EVEN"

ELSE

PRINT "ODD"

END IF

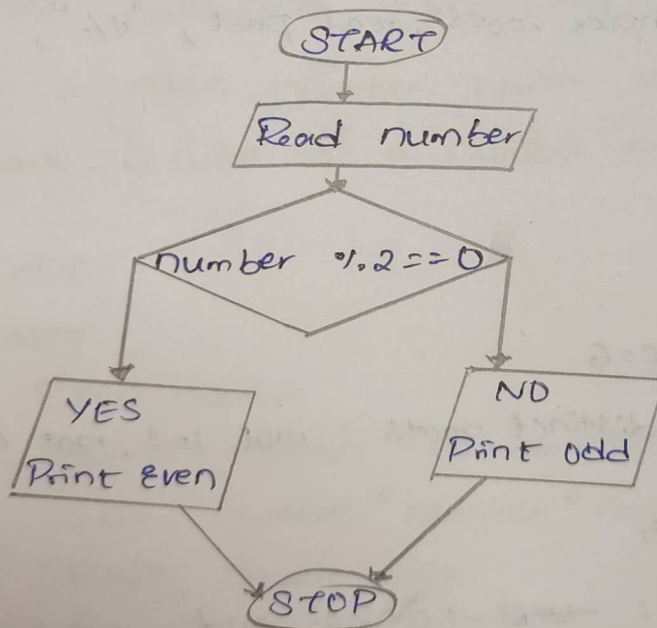
STOP

INPUT/OUTPUT:

i/p number = 24 → O/p = Even

i/p number = 17 → O/p = Odd

FLOW CHART:



⑧ An Engineer needs to Solve quadratic eqns of the form.
Write a C program to Calculate and display the roots
of the quadratic equation.

PSEUDO CODE:

START

READ a, b, c

$D = b^2 - 4 * a * c$

IF $b > 0$ THEN

root 1 = $(-b + \text{Sqrt}(D)) / (2 * a)$

root 2 = $(-b - \text{Sqrt}(D)) / (2 * a)$

PRINT "Real and distinct roots", root 1, root 2

ELSE IF $D = 0$ THEN

root 1 = $-b / (2 * a)$

PRINT "Real and equal roots", root 1

ELSE

realPart = $-b / (2 * a)$

imagPart = $\text{Sqrt}(-D) / (2 * a)$

PRINT "Complex roots", real part, "+/-",

imagPart, "i"

END IF

STOP

INPUT/OUTPUT:

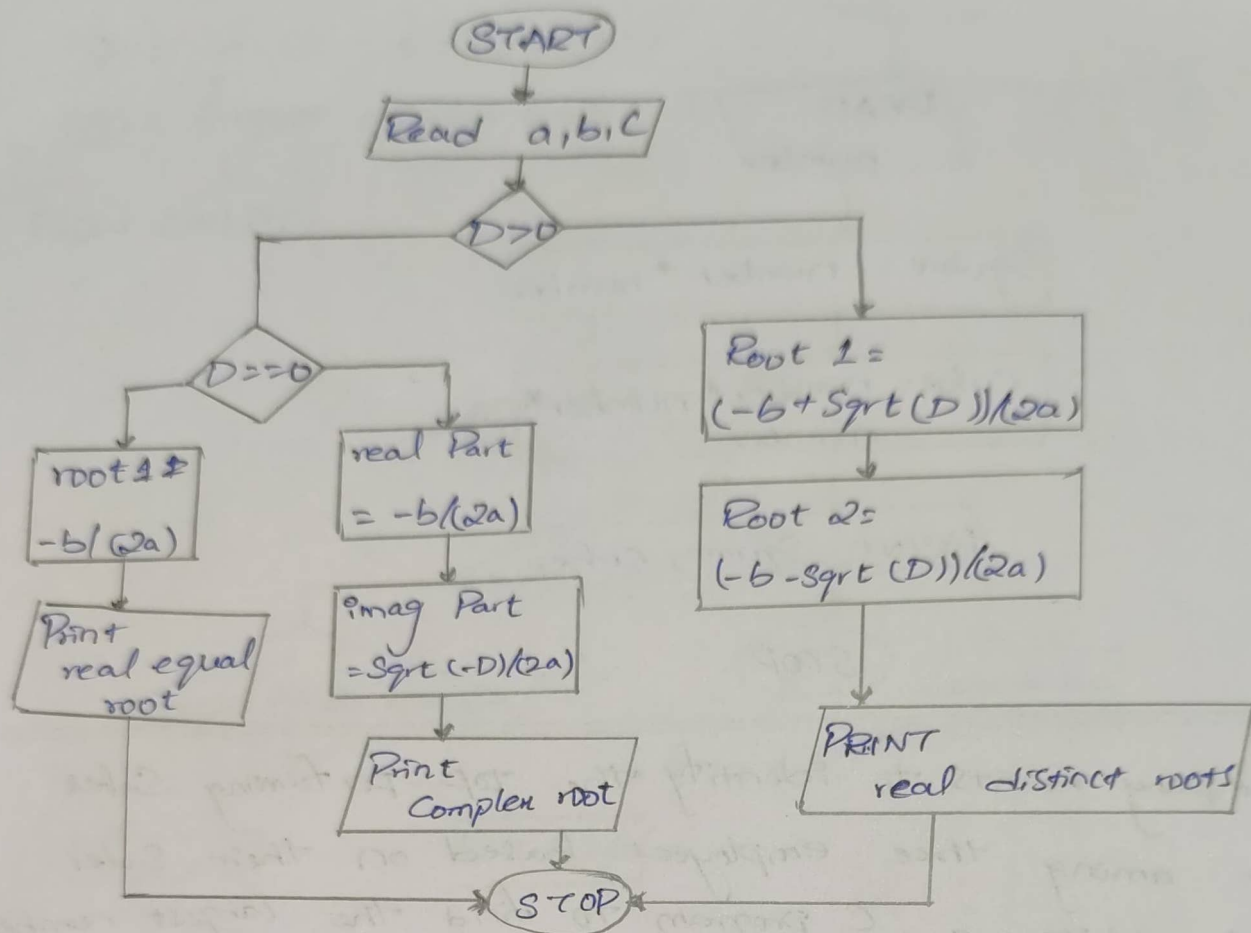
Input: $a=1, b=-5, c=6$

Output: Real and distinct roots: root 1=3, root 2=2

Input: $a=1, b=2, c=5$

Output: Complex roots: ~~-1.0~~ -1.0+2.0i and -1.0-2.0i.

FLOW CHART:



⑨ A Scientific Calculator application requires the Square and Cube of a given number. Write a C program to Calculate the Square & Cube of a number entered by the user.

PSEUDO CODE:

START

READ number

Square = number * number

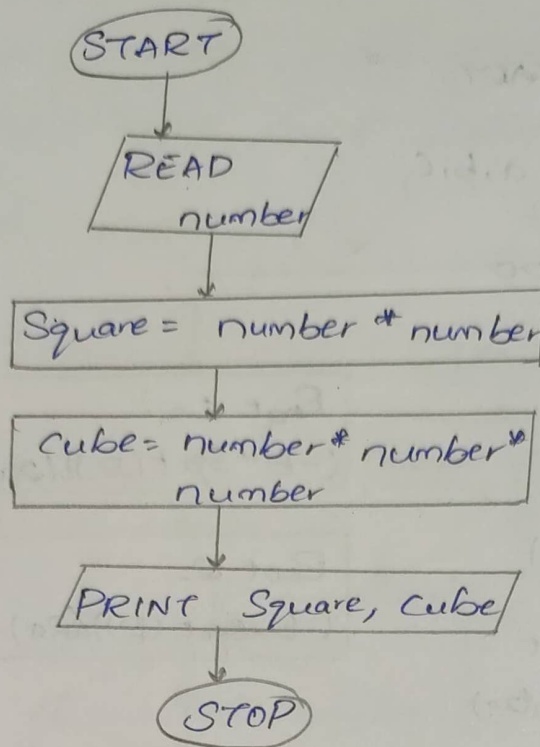
Cube = number * number * number

PRINT Square, Cube

STOP

INPUT / OUTPUT:

number = 4 Output: Square = 16 , Cube = 64



- ⑩ A Company wants to identify the top-performing Sales Person among three employees based on their Sales figures. Write a C program to find the largest number among three numbers.

PSEUDO CODE:

START

READ a, b, c

IF $a \geq b$ AND $a \geq c$ THEN

largest = a

ELSE IF $b \geq a$ AND $b \geq c$ THEN

largest = b

ELSE

largest = c

END IF

PRINT largest

STOP

INPUT / OUTPUT:

I/P : $a=20$, $b=45$, $c=33$

O/P : largest number = 45

FLOW CHART:

